

# Nanoplastics in the human brain and their change in abundance over time

Nanoplastics were identified in human brain samples from cadavers. Concentrations were higher in the brain than in the liver or kidneys, and appeared to increase over time between 2016 and 2024. Visualization of the plastics using a variety of approaches provided insights into the anatomical distribution and physical characteristics of nanoplastics.

## This is a summary of:

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## The question

Microplastics have been identified in various tissues of the human body, representing an important global health issue<sup>1</sup>. However, as conventional detection methods often exclude material smaller than 5  $\mu\text{m}$ , nanoplastics (1 to 1000 nm) are frequently unaccounted for in quantitative assessments. This detection gap confounds the interpretation of in vitro and in vivo toxicological studies that investigate the controlled exposure of microplastics, as the abundance, type, size and shape of nanoplastics already in the human body are essentially unknown. Using novel quantification techniques, we explored the presence of micro- and nanoplastics (MNPs) in human tissues, with the aim to bridge this critical knowledge gap and provide insights into potential exposure and health risks associated with these pervasive environmental contaminants. To achieve this, we required a quantitative approach that included all sizes of plastic particles.

## The observation

Our study utilized de-identified postmortem human brain, liver and kidney samples obtained from autopsies conducted in 2016 and 2024 in New Mexico. Using novel pyrolysis gas chromatography–mass spectroscopy (Py-GC/MS) techniques<sup>2,3</sup>, we identified significantly higher concentrations of MNPs in the brain than in the liver and kidneys. Polyethylene was found to be the most abundant polymer, followed by polypropylene, polyvinyl chloride and styrene-butadiene rubber. Further analysis revealed that samples obtained in 2024 had approximately 50% higher MNP concentrations in brain tissue than samples from 2016 (Fig. 1a), consistent with rising environmental MNP levels. Brain samples from earlier timepoints (1997–2013), though geographically dissimilar, also showed increasing levels of MNPs over time. Finally, samples from cadavers of people with dementia demonstrated higher plastic concentrations than those without dementia, potentially linked to impaired blood–brain barrier integrity and reduced clearance mechanisms, which are distinctive features of neurodegenerative conditions.

We next used several other approaches to confirm our Py-GC/MS data. Scanning electron microscopy (SEM) and polarization wave microscopy confirmed

the presence of refractile inclusions in tissues. Moreover, transmission electron microscopy (TEM) provided detailed visualization of isolated particles, with sizes in the nanoscale range, typically between 100 and 200 nm (Fig. 1b).

## The implications

It was more than 20 years ago that the term microplastics was first introduced<sup>4</sup>. However, it is only within the past 5 years that scientists have begun to observe MNPs in the human body. Our findings suggest that, rather concerning, MNP concentrations in humans might have more than doubled in the time span between their first ecological detection and now.

The discovery of MNPs in human brain tissue, and the trend of increased MNP abundance over time, should provide justification for new public health and environmental policies around the production and waste management of plastics. The high abundance of MNPs in the brains of cadavers from people with dementia points to a possible link between impaired brain health and increased plastic accumulation. However, an alternative explanation is that nanoplastics accumulate because of a disease typified by an impaired blood–brain barrier, atrophy and inflammation. Understanding how MNPs enter the human body is crucial for developing effective strategies to mitigate exposure. It is not known if current concentrations increase risk of disease, nor do our studies indicate any threshold for disease incidence in the future.

The prevalence of nanoscale plastics, often undetected by traditional methods, reinforces the need for advanced methodologies and monitoring to accurately assess MNP exposure in both environmental and biological settings. Based on imaging findings, we propose that MNPs in the human body are very old, flaked residues of plastics discarded long ago, although further experiments are needed to confirm their source. Much of the public messaging for avoiding MNPs emphasizes exposures from fresh plastics (such as water bottles and cutting boards), and so it is crucial to understand whether these are indeed the major sources of MNPs to ensure we intervene effectively to mitigate future problems for human and ecological health.

**Marcus Garcia & Matthew Campen**

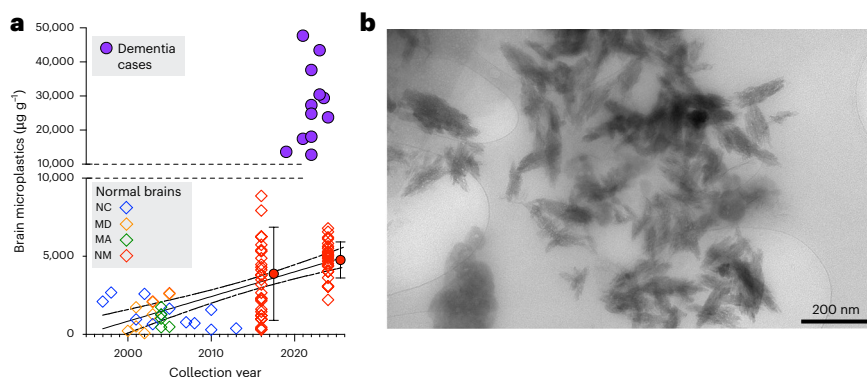
University of New Mexico College of Pharmacy, Albuquerque, NM, USA.

## EXPERT OPINION

"The manuscript deals with a very current and concerning societal issue: the possible risk due to the very high levels of plastics that we could be exposed to. Environmental plastics are not biodegradable (hence, can bioaccumulate), and the daily use of plastics is increasing without increasing recycling. Regardless of how they enter the brain,

finding microplastics in the brain is highly interesting because it means that plastics can cross the blood–brain barrier. It will support some current hypotheses relating mental health to environmental pollution." **Jorge Bernardino de la Serna, Imperial College London, London, UK.**

## FIGURE



**Fig. 1 | Nanoplastics in the brain.** **a**, Py-GC/MS quantitation of plastics in the prefrontal cortex from human cadaver samples, with or without diagnosed dementia. NC,  $n = 13$  samples obtained from Duke Kathleen Price Bryan Brain Bank; MD,  $n = 9$  samples obtained from the Harvard Brain Tissue Resource Center; MA,  $n = 5$  samples obtained from the NICHD Brain and Tissue Bank at the University of Maryland; NM,  $n = 52$  samples obtained from the University of New Mexico. **b**, Example TEM image showing shard- or flake-like solid particulates, the majority of which being  $<200$  nm in length and  $<40$  nm in width. © 2025, Nihart, A. J. et al., CC BY-NC-ND 4.0.

## BEHIND THE PAPER

During one of our weekly lab meetings, we were discussing polarization wave images that revealed clear refractile inclusions in liver and kidney that were in the 1–5  $\mu\text{m}$  range. However, the brain images did not reveal such large particles, and instead showed submicron particles that were infrequent, seemingly at odds with the Py-GC/MS data. Based on these data, we made the bold assumption that the brain contained plastics that were on

the nanoscale and simply not visible with light microscopy, although we were later challenged by Eliane El Hayek, who stated that "If we are going to say nanoplastics, we need to see nanoplastics". This statement became a guiding principle, leading to us combining multiple complementary tools such as TEM, SEM, Py-GC/MS and polarization wave microscopy to fully appreciate the nature of MNPs in the brain. **M.G. & M.C.**

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## FROM THE EDITOR

"This study is critical given the exponential rise in global concentrations of environmental micro- and nanoplastics. The findings suggest that plastic nanoparticles selectively accumulate in the human brain and that internal concentrations have increased from 2016 to 2024." **Editorial Team, Nature Medicine.**